ALPHAEUS DMONTE

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SUMMARY

PhD candidate at George Mason University, specializing in Natural Language Processing and Large Language Models. Experienced in fine-tuning LLMs, developing scalable NLP solutions, and building multilingual datasets, with publications at leading conferences and workshops. Skilled in LLM post-training, text understanding and generation. Proven industry experience and seeking a Research/Applied Scientist opportunity to apply NLP expertise to real-world challenges

RELEVANT EXPERIENCE

Cambium Assessment *Remote*
Machine Learning Intern *June 2026 - Present*

- Implemented training pipeline using transformer based language models for Spanish Automated Scoring achieving 80% accuracy
- Researched approaches for improving the performance of domain specific scoring systems for Science essays, and developed translation-based approaches for model training and evaluation
- Developed data augmentation techniques to reduce class imbalance in model training, improving accuracy by 5%

George Mason University *Fairfax, VA*
Graduate Research Assistant *January 2023 – Present*

- Architected SUNDER to improve contextual understanding in LLMs, delivering F1 score gains of up to 15% over baseline models
- Implemented an LLM-driven question-answering approach leveraging RAG, yielding a 3% accuracy gain and improving computational efficiency by 33% over baselines
- Developed a novel multilingual dataset and an IR-based lexical simplification pipeline using transformer models, achieving 76.5% accuracy
- Fine-tuned and benchmarked LLMs for NLP tasks, including text classification, span detection, and question answering improving baseline performance upto 3%.
- Created and open-sourced benchmark datasets of human, AI, and hybrid text and evaluated language models for text characterization and authorship attribution, achieving an F1 score of 98%

Qualtrics *Reston, VA*
Applied Scientist Intern *June 2025 - August 2025*

- Analyzed and applied model merging methods in a multilingual setting, improving cross-lingual performance across 5 languages
- Designed and executed experiments across multiple tasks, benchmarking merging strategies and achieving performance gains of up to 5%
- Developed and automated evaluation pipelines, reducing training time by 67% and compute costs by 60%

Retail Business Services *Quincy, MA*
Machine Learning/Data Science Co-op *January 2022 – June 2022*

- Developed an intelligent reporting system using Rasa to query PowerBI dashboards and data sources, reducing data query time by 50% for business analysts
- Fine-tuned computer vision models using PyTorch on Azure Percept DK, achieving 94% accuracy in product detection for inventory management
- Implemented automated end-to-end ML pipelines using Azure ML and GitHub actions, streamlining model training and deployment

SKILLS

- **Programming Languages:** Python, Java, C, C++, SQL, Javascript,
- **Databases:** PostgreSQL, MongoDB
- **ML Frameworks & Libraries:** Transformers, Huggingface, LangChain, PyTorch, TensorFlow, Scikit-learn, Pandas, Numpy
- **Cloud Platforms & Tools:** Git, Google Colab, Databricks, Azure ML, AWS, Docker
- **Web Development:** Flask, React

EDUCATION

PhD, Information Technology *Expected May 2027*
George Mason University *Fairfax, VA*

- Advisor: Dr. Marcos Zampieri

MS, Computer Science *December 2022*
Rochester Institute of Technology *Rochester, NY*

BE, Computer Engineering *October 2020*
University of Mumbai *Mumbai, India*

PUBLICATIONS

Dmonte, A., Ranasinghe, T., Zampieri, M., 2026. SUNDER: Selective Unmasking for Text Understanding. *Accepted at EMNLP*, 2026

Dmonte, A., Gupta, V., Perry, D., Arehart, M., 2026. Improving Training Efficiency and Reducing Maintenance Costs via Language Specific Model Merging. *In Proceedings of EACL*, 2026

Dmonte, A., Oruche, R., Ranasinghe, T., Zampieri, M., Calyam, P., 2025. Exploring the Performance of Large Language Models on Subjective Span Identification Tasks. *In Proceedings of AACL-IJCNLP*, 2025

Oruche, R., **Dmonte, A.**, Seth, V., Zeng, Z., Zhang, Y., Zampieri, M., Calyam, P. Disentangling Complex Questions in LLMs via Multi-Hop Dependency Graphs. *In Proceedings of CIKM*, 2025

Dmonte, A., Satapara, S., Alsudais, R., Ranasinghe, T., Zampieri, M. On the Effects of Machine Translation on Offensive Language Detection. *In Social Network Analysis and Mining*, 2025

Dmonte, A., Ko, E., Zampieri, M. An Evaluation of Large Language Models in Financial Sentiment Analysis. *In Proceedings of IEEE International Workshop on Large Language Models for Finance*, 2024

Dmonte, A., Arya, T., Ranasinghe, T., Zampieri, M. Towards Generalized Offensive Language Identification. *In Proceedings of ASONAM*, 2024

Dmonte, A., Zampieri, M., Lybarger, K., Albanese, M. and Coulter, G. Classifying Human-Generated and AI-Generated Election Claims in Social Media. *In Proceedings of SECRYPT*, 2024

Zampieri, M., Rosenthal, S., Nakov, P., **Dmonte, A.**, Ranasinghe, T. OffensEval 2023: Offensive language identification in the age of Large Language Models. *In Natural Language Engineering*, 2023

North, K., **Dmonte, A.**, Ranasinghe, T., Shardlow, M., Zampieri, M. ALEXSIS+: Improving substitute generation and selection for lexical simplification with information retrieval. *In Proceedings of the 18th Workshop on Innovative Use of NLP for Building Educational Applications*, 2023

CONFERENCE PRESENTATIONS

- EACL 2026
- AACL-IJCNLP 2025
- ASONAM 2024
- Workshop on Large Language Models for Finance 2024
- CCI Symposium 2024

AWARDS

- Doctoral Research Scholars Award, *George Mason University*
- Merit-based Scholarship, *Rochester Institute of Technology*